

Angle Rules and Polygons

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1

[1 mark]

Two angles lie on a straight line. One is 115° .

Work out the other angle.

1 Angles on a straight line add up to 180° .

2 $180 - 115 = 65$

Answer: 65°

2

[2 marks]

Three angles meet at a point. Two of them are 90° and 130° .

Work out the third angle.

1 Angles around a point add up to 360° .

2 $90 + 130 = 220$

3 $360 - 220 = 140$

Answer: 140°

3

[2 marks]

Two straight lines cross. One of the angles formed is 72° .

(a) Write down the angle vertically opposite it.

(b) Work out one of the other two angles.

1 (a) Vertically opposite angles are equal, so it is 72° .

2 (b) The other angles lie on a straight line with the 72° angle.

3 $180 - 72 = 108^\circ$

Answer: (a) 72° (b) 108°

4

[2 marks]

A triangle has angles of 43° and 90° .

Work out the third angle.

1 The angles of a triangle add up to 180° .

2 $43 + 90 = 133$

3 $180 - 133 = 47$

Answer: 47°

5

[3 marks]

An isosceles triangle has an apex angle of 40° .

Work out the size of each of the two base angles.

- 1 In an isosceles triangle the two base angles are equal.
- 2 The three angles total 180° , so the two base angles share $180 - 40 = 140^\circ$.
- 3 $140 \div 2 = 70^\circ$ each.

Answer: 70° each

6

[3 marks]

A quadrilateral has angles of 85° , 100° , 120° and x° .

Work out the value of x .

- 1 The angles of a quadrilateral add up to 360° .
- 2 $85 + 100 + 120 = 305$
- 3 $x = 360 - 305 = 55$

Answer: $x = 55$

7

[3 marks]

(a) Work out the size of one exterior angle of a regular pentagon.

(b) Work out the size of one interior angle of a regular pentagon.

- 1 **(a)** The exterior angles of any polygon add up to 360° .
- 2 For a regular pentagon: $360 \div 5 = 72^\circ$
- 3 **(b)** An interior angle and its exterior angle lie on a straight line.
- 4 $180 - 72 = 108^\circ$

Answer: (a) 72° (b) 108°

8

[3 marks]

Two parallel lines are crossed by a straight line. One of the angles formed at the first crossing is 118° .

(a) Write down the size of the corresponding angle at the second crossing, with a reason.

(b) Work out the size of the co-interior angle, with a reason.

- 1 **(a)** Corresponding angles (in the same position at each crossing) are equal on parallel lines.
- 2 So the corresponding angle is 118° .
- 3 **(b)** Co-interior angles lie between the parallel lines on the same side of the crossing line and add up to 180° .
- 4 $180 - 118 = 62^\circ$

Answer: (a) 118° (corresponding angles are equal) (b) 62° (co-interior angles add to 180°)

9

[4 marks]

- (a) Work out the sum of the interior angles of a hexagon.
 (b) Five of the angles of a hexagon are 100° , 120° , 130° , 140° and 110° . Work out the sixth angle.

1 (a) Sum of interior angles = $(n - 2) \times 180^\circ$ with $n = 6$

2 $= 4 \times 180 = 720^\circ$

3 (b) Add the five known angles: $100 + 120 + 130 + 140 + 110 = 600$

4 $720 - 600 = 120^\circ$

Answer: (a) 720° (b) 120°

10

[4 marks]

A regular polygon has an exterior angle of 24° .

- (a) How many sides does it have?
 (b) Work out one interior angle.
 (c) Work out the sum of its interior angles.

1 (a) The exterior angles total 360° , so the number of sides is $360 \div 24 = 15$

2 (b) Interior angle = $180 - 24 = 156^\circ$

3 (c) Sum = $(15 - 2) \times 180 = 13 \times 180 = 2340^\circ$

4 Check: $15 \times 156 = 2340 \checkmark$

Answer: (a) 15 sides (b) 156° (c) 2340°

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Work out the sum of the interior angles of a decagon (10 sides).

1 Sum = $(n - 2) \times 180^\circ = (10 - 2) \times 180$

2 $= 8 \times 180 = 1440^\circ$

Answer: 1440°

2

[2 marks]

A regular polygon has an interior angle of 162° .

How many sides does it have?

1 Exterior angle = $180 - 162 = 18^\circ$

2 Number of sides = $360 \div 18 = 20$

Answer: 20 sides

3

[3 marks]

In a regular polygon, each interior angle is 4 times the size of each exterior angle.
How many sides does the polygon have?

- 1 Let the exterior angle be e , so the interior angle is $4e$.
- 2 They lie on a straight line: $e + 4e = 180$
- 3 $5e = 180$, so $e = 36^\circ$
- 4 Number of sides = $360 \div 36 = 10$

Answer: 10 sides

4

[3 marks]

The angles of a triangle are $(3x - 10)^\circ$, $(2x + 20)^\circ$ and $(x + 50)^\circ$.
Work out the size of each angle.

- 1 Add them and set equal to 180: $3x + 2x + x = 6x$ and $-10 + 20 + 50 = 60$
- 2 $6x + 60 = 180$, so $6x = 120$ and $x = 20$
- 3 The angles are $3(20) - 10 = 50^\circ$, $2(20) + 20 = 60^\circ$ and $20 + 50 = 70^\circ$
- 4 Check: $50 + 60 + 70 = 180$ ✓

Answer: 50° , 60° and 70°

5

[4 marks]

Prove that the exterior angle of a triangle is equal to the sum of the two interior angles at the other two vertices.

- 1 Call the three interior angles a , b and c , and let the exterior angle at the c vertex be e .
- 2 The angles of a triangle add up to 180° , so $a + b + c = 180$
- 3 The interior angle c and the exterior angle e lie on a straight line, so $c + e = 180$
- 4 From the second equation, $e = 180 - c$
- 5 From the first equation, $a + b = 180 - c$
- 6 Both expressions equal $180 - c$, so $e = a + b$ as required.

Answer: $e = 180 - c = a + b$

6

[4 marks]

A regular hexagon, a square and one other regular polygon fit together exactly around a point, with no gaps.

How many sides does the third polygon have?

- 1 Interior angle of a regular hexagon = $180 - (360 \div 6) = 120^\circ$
- 2 Interior angle of a square = 90°
- 3 Angles around a point total 360° , so the third angle is $360 - 120 - 90 = 150^\circ$
- 4 Its exterior angle is $180 - 150 = 30^\circ$
- 5 Number of sides = $360 \div 30 = 12$

Answer: 12 sides

7

[4 marks]

A regular octagon and a regular pentagon are placed so that they share one complete side. Work out the size of the angle between them at one of the shared vertices, measured outside both shapes.

- 1 Interior angle of a regular octagon = $180 - (360 \div 8) = 180 - 45 = 135^\circ$
- 2 Interior angle of a regular pentagon = $180 - (360 \div 5) = 180 - 72 = 108^\circ$
- 3 At the shared vertex, the two interior angles and the outside angle fit around a point.
- 4 Outside angle = $360 - 135 - 108$
- 5 = 117°

Answer: 117°

8

[4 marks]

The five interior angles of a pentagon are x° , $(x + 20)^\circ$, $(x + 30)^\circ$, $(x + 40)^\circ$ and $(x + 50)^\circ$. Work out the size of the largest angle.

- 1 Sum of the interior angles of a pentagon = $(5 - 2) \times 180 = 540^\circ$
- 2 Adding the expressions: $5x + (20 + 30 + 40 + 50) = 5x + 140$
- 3 $5x + 140 = 540$, so $5x = 400$ and $x = 80$
- 4 The largest angle is $x + 50 = 130^\circ$
- 5 Check: $80 + 100 + 110 + 120 + 130 = 540 \checkmark$

Answer: 130°

9

[4 marks]

Prove, using the properties of parallel lines, that the angles of a triangle add up to 180° .

- 1 Take triangle ABC with interior angles a at A, b at B and c at C.
- 2 Draw a line through C parallel to AB.
- 3 The angle between that line and CA (on the A side) equals a , because alternate angles on parallel lines are equal.
- 4 Likewise the angle between the line and CB equals b , again by alternate angles.
- 5 Those two angles together with c lie on a straight line through C, so they add up to 180° .
- 6 Therefore $a + b + c = 180^\circ$.

Answer: Drawing a parallel line through one vertex gives two alternate angles equal to a and b , and $a + b + c$ is then a straight line = 180°

In a regular polygon, each interior angle is 20° more than 7 times each exterior angle.

(a) How many sides does the polygon have?

(b) Work out the size of one interior angle.

(c) Work out the sum of the interior angles.

1 **(a)** Let the exterior angle be e , so the interior angle is $7e + 20$.

2 Interior and exterior angles lie on a straight line: $e + 7e + 20 = 180$

3 $8e = 160$, so $e = 20^\circ$

4 Number of sides = $360 \div 20 = 18$

5 **(b)** Interior angle = $180 - 20 = 160^\circ$ (check: $7 \times 20 + 20 = 160$ ✓)

6 **(c)** Sum = $(18 - 2) \times 180 = 16 \times 180 = 2880^\circ$

Answer: (a) 18 sides (b) 160° (c) 2880°